

Module 5

Two-Port Networks and Network Graphs

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Transmission parameters (ABCD-parameters)

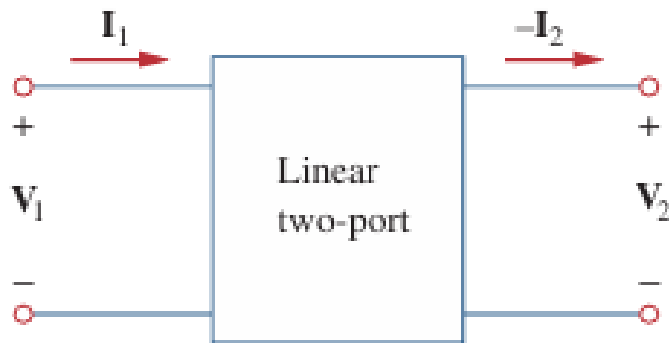
Set of parameters which relates the variables at the input port to those at the output port.

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{A}\mathbf{V}_2 - \mathbf{B}\mathbf{I}_2 \\ \mathbf{I}_1 &= \mathbf{C}\mathbf{V}_2 - \mathbf{D}\mathbf{I}_2 \end{aligned}$$

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} = [\mathbf{T}] \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix}$$

The two-port parameters in the equations provide a measure of how a circuit transmits voltage and current from a source to a load. They express sending-end variables (V_1 , I_1) in terms of the receiving-end variables (V_2 and I_2). For this reason, they are called **transmission parameters**.

They are also known as ABCD parameters. They are used in the design of telephone systems, microwave networks, and radars.



Transmission parameters (ABCD-parameters)

The transmission parameters are determined as

$$\mathbf{A} = \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, \quad \mathbf{B} = - \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0}$$
$$\mathbf{C} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, \quad \mathbf{D} = - \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0}$$

A = Open-circuit voltage ratio

B = Negative short-circuit transfer impedance

C = Open-circuit transfer admittance

D = Negative short-circuit current ratio

A and D are dimensionless, B is in ohms, and C is in siemens. Since the transmission parameters provide a direct relationship between input and output variables, they are very useful in cascaded networks.

Transmission parameters (abcd-parameters)

Another set of parameters may be defined by expressing the variables at the output port in terms of the variables at the input port.

$$\begin{aligned} V_2 &= aV_1 - bI_1 \\ I_2 &= cV_1 - dI_1 \end{aligned}$$

$$\begin{bmatrix} V_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} V_1 \\ -I_1 \end{bmatrix} = [t] \begin{bmatrix} V_1 \\ -I_1 \end{bmatrix}$$

The parameters a, b, c, d are called the inverse transmission parameters.

$$\begin{aligned} a &= \left. \frac{V_2}{V_1} \right|_{I_1=0}, & b &= - \left. \frac{V_2}{I_1} \right|_{V_1=0} \\ c &= \left. \frac{I_2}{V_1} \right|_{I_1=0}, & d &= - \left. \frac{I_2}{I_1} \right|_{V_1=0} \end{aligned}$$

a = Open-circuit voltage gain

b = Negative short-circuit transfer impedance

c = Open-circuit transfer admittance

d = Negative short-circuit current gain

While a and d are dimensionless, b and c are in ohms and siemens, respectively.

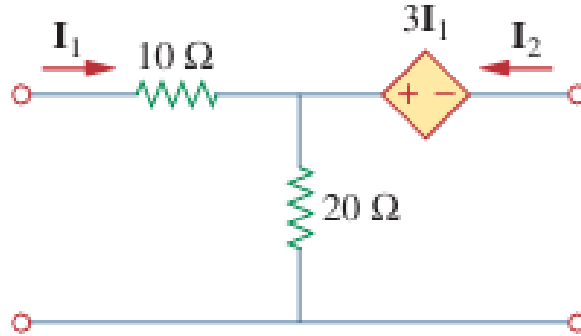
In terms of the transmission or inverse transmission parameters, a network is **reciprocal** if

$$AD - BC = 1, \quad ad - bc = 1$$

and **symmetrical** if $A = D$

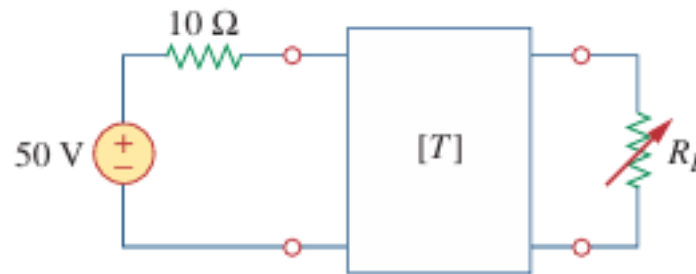
Transmission parameters

1. Find the transmission parameters for the two-port network.



2 The ABCD parameters of the two-port network are given below. The output port is connected to a variable load for maximum power transfer. Find R_L and the maximum power transferred.

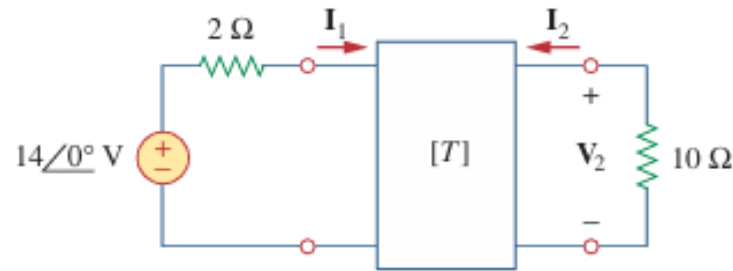
$$\begin{bmatrix} 4 & 20\ \Omega \\ 0.1\ \text{S} & 2 \end{bmatrix}$$



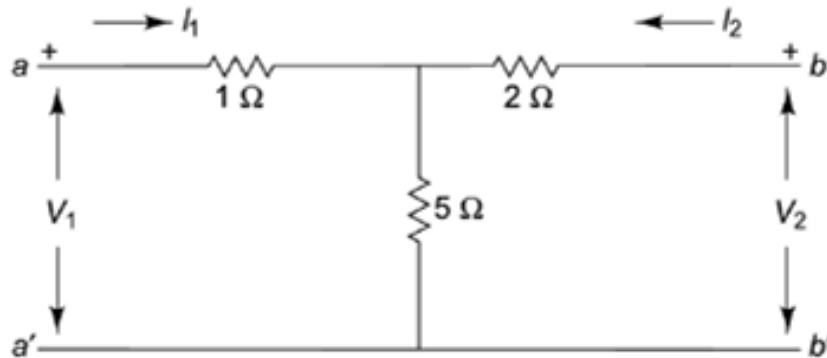
Transmission parameters

3. Find I_1 and I_2 if the transmission parameters for the two-port network is as given below.

$$\begin{bmatrix} 5 & 10 \Omega \\ 0.4 \text{ S} & 1 \end{bmatrix}$$

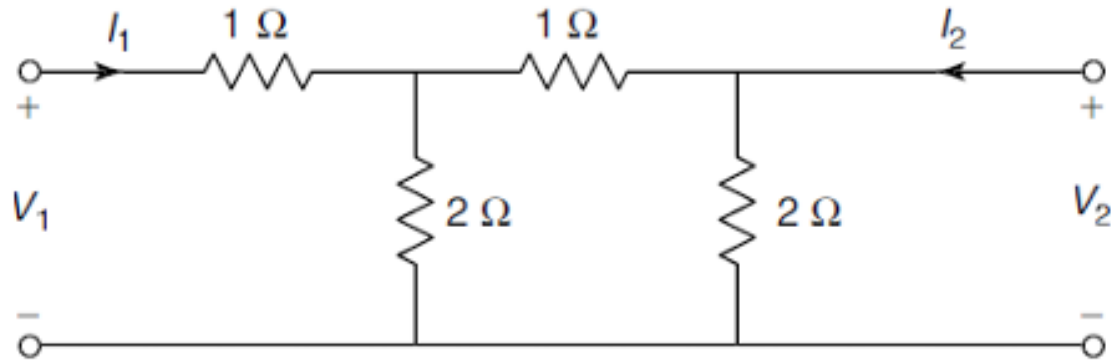


4. Find the ABCD parameters for the network.

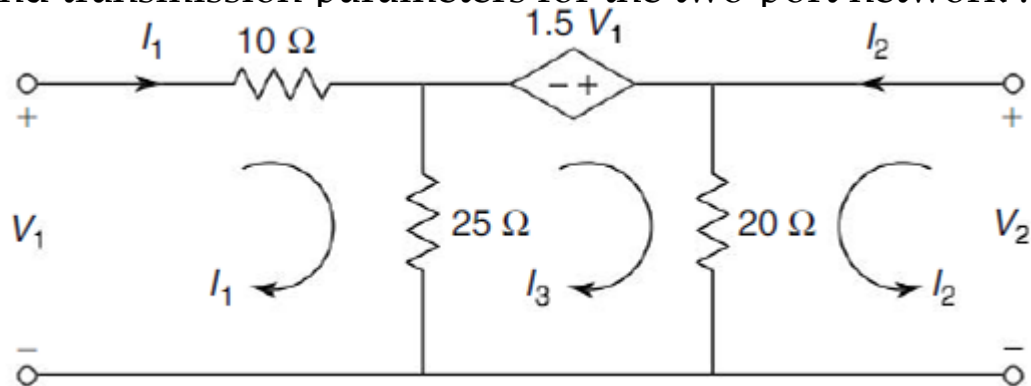


Transmission parameters

5. Obtain ABCD parameters for the network.



6. Find transmission parameters for the two-port network .



Next class: Relationship between parameters